Texture Feature: Variance

In texture segmentation, variance is a crucial metric that helps us understand how much the pixel intensities in a local neighborhood deviate from the local average.

$$Variance = Mean(X^2) - [Mean(X)]^2$$

For our example:

If we apply this to a specific window of pixels (represented by X), the formula dictates that we:

1. Square all the pixel values in the window and find their mean: $Mean(X^2)$.
2. Find the mean of the original pixel values, and then square that result: $[Mean(X)]^2$.
3. Subtract the latter from the former to obtain the local variance.

A higher variance indicates a rough or complex texture, while a lower variance points to a smooth texture.